
A Structural Load Model Cannot Predict the Effect of a Conservation Appeal

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Abstract

Temporal models are increasingly asked to answer interventional queries rather than only forecast. We report a case study on what happens when one is held to that standard. We decompose CAISO system load over 2016 to 2024 into a thermal layer, hour-of-day behavioral coefficients for conservation appeals and pandemic occupancy identified by matched difference-in-differences, and a gradient-boosted residual, then subject it to four holdouts. On the 2023 to 2024 accuracy holdout, coefficients from a careful matched design and from naive full-series regression differ by 0.04 MAPE points, so the forecasting metric cannot rank the two designs even though their implied appeal effects differ twofold. On an interventional holdout of the ten consecutive Flex Alert days of the September 2022 California heat wave, every model predicts a nearly constant appeal effect of one to two percentage points while the observed effect ranges from +7.4 to −19.9. No model separates from a baseline predicting no effect at all, both structural variants are nominally worse than it, and the identified model’s per-day ranking is significantly anti-correlated with the truth ($\rho_s = -0.76$, $p = 0.011$). A held-out lockdown regime is the counterweight: removing the occupancy term costs 4.9 MAPE points, so structure carries information where identification does not. We evaluate a task-specific structural model rather than a pretrained temporal foundation model, and are explicit about what that supports. The shift that defeats the model is in the treatment rather than the covariates, so leakage-aware splits cannot see it.

1 Introduction

The case for putting structure into a temporal model is that structure buys counterfactuals. A pure forecaster says what demand will be; a model with a thermal layer and a behavioral layer promises to say what demand *would have been* three degrees warmer, or without the conservation appeal. That capability is what decides whether a model can support a decision.

Such models are validated with the wrong test. A counterfactual load model is graded on held-out MAPE, pinball loss and coverage [Hong and Fan, 2016, Hong et al., 2016]. Those metrics score the factual prediction, are computed on days where no intervention occurred, and are silent about whether the behavioral layer means anything [Shmueli, 2010]. A model can win the accuracy benchmark while its appeal coefficient is confounded, mis-signed, or a constant that ignores the day.

Scope. We evaluate a task-specific structural model: a hand-specified calendar and thermal equation, hour-of-day multiplicative coefficients for two event families, and a gradient-boosted residual (Appendix A specifies it fully). We do not evaluate a pretrained temporal foundation model, and nothing here measures how one would behave. Our evidence supports something narrower: for a

Research conducted at AIscent Labs, <https://aiscent-research.github.io>.

temporal model whose interventional claims are explicit and inspectable, standard forecasting metrics were uninformative about those claims, and the failure we isolate lies in how the treatment was represented rather than in the architecture. A model that represents interventions implicitly is harder to audit this way, which is why we report a protocol rather than a result about a model class.

Findings. On the accuracy holdout, behavioral coefficients from a careful matched design and from naive full-series regression are separated by 0.04 MAPE points, and both lose to a plain gradient-boosted model with no structure, so the forecasting metric is blind to the difference. On an interventional holdout of ten consecutive Flex Alert days from the September 2022 California heat wave, every model predicts a nearly constant appeal effect of one to two percentage points while the observed effect ranges from +7.4 to −19.9; no model separates from a zero-effect baseline, and the identified model’s ordering across days is significantly anti-correlated with the truth ($\rho_s = -0.76$, $p = 0.011$). A held-out lockdown regime is the counterweight: removing the occupancy term costs 4.9 MAPE points, so structure does real work where identification does not.

What is new here. That predictive accuracy does not establish causal validity is long established [Shmueli, 2010, Pearl, 2009], and we do not claim it. We contribute four things: a measurement of how uninformative the forecasting metric is, where two designs whose causal claims differ twofold are separated by 0.04 MAPE points; a zero-effect comparator reported as a table row, which makes the null legible and is not standard in load modelling; isolation of *treatment-label non-stationarity* as the operative failure, where covariates stay in support and the label is stable while the intervention changes; and a ground-truth-free trust region on the query.

2 Model, identification, and holdouts

We use hourly CAISO system load from 2016-01-01 to 2024-12-31 (3,287 days) with hourly temperature and dew point, in three layers. A **thermal** layer maps a smoothed afternoon temperature index ($\alpha = 0.847$) through a piecewise response with cooling threshold $T_c = 17.65^\circ\text{C}$ ($b_c = 1056$, $b_{c2} = 43.0$ MW) and heating threshold $T_h = 13.23^\circ\text{C}$ ($b_h = 257$ MW), over a day-type, day-of-year and year baseline. A **behavioral** layer adds hour-of-day log-multiplicative coefficients for 43 CAISO Flex Alert days (2016 to 2022), constrained to the appeal window, and 89 pandemic occupancy days (2020-03-19 to 2020-06-15) interacted with day type. A **residual** layer of gradient-boosted trees [Ke et al., 2017] at the 10th, 50th and 90th percentiles models the remainder. Appendix A gives the functional form, feature list and training protocol.

The *identified* variant estimates each behavioral coefficient by matched difference-in-differences, pooling the hourly log-ratio contrast between each event day and control days matched on temperature and calendar [Imbens and Rubin, 2015, Callaway and Sant’Anna, 2021]. The *ablation* keeps the identical architecture and fits the same coefficients by full-series regression on an event indicator, which is what a practitioner gets without a control-day design. The two agree on every thermal parameter to within 2 percent and disagree on the peak appeal coefficient by a factor of two (+3.2 against +6.4 percent). Baselines are a *flag-as-feature* model that receives the event indicators as ordinary features and answers counterfactuals by toggling the flag, a *plain* model that never sees them and therefore implies an effect of exactly zero, 168-hour persistence, and k -nearest-neighbour similar-day.

Four holdouts. Nothing in any holdout enters any stage of fitting. **(A) Accuracy:** 2023 and 2024 (730 days), scored by MAPE, pinball loss and coverage [Gneiting and Raftery, 2007]. **(B) Interventional:** the ten consecutive Flex Alert days of the September 2022 California heat wave, 2022-08-31 to 2022-09-09 [California Public Advocates Office, 2022], with the other 33 alert days, all from 2016 to 2021, left in training. For each held-out day we compute an observed matched difference-in-differences effect over the 4 to 9 p.m. window by the procedure that produced the training estimates, and score each model’s implied effect against it in percentage points of effect. **(C) Regime shift:** held-out lockdown weekdays, 2020-05-01 to 2020-06-15 (46 days), plus two structure-only variants separating the value of *having* an occupancy term from that of estimating it carefully. **(D) External replication:** the Low Carbon London dynamic time-of-use trial [Schofield et al., 2015], in which 1,122 of 5,567 metered households received a dynamic tariff through 2013; we identify the price response on a random half of treated households and predict the other half. **(E) Analog agreement:** we retrieve the twelve nearest analog days and ask what fraction of counterfactual hours

(A) Accuracy holdout, 2023 to 2024				(B) Interventional holdout, 10 days			
Model	MAPE	Pinball ₅₀	Cov ₈₀	Model	Bias	MAE	ρ_s
Identified	3.79	479	64.9	Identified	+3.69	6.36	−0.76*
Ablation	3.75	474	65.8	Ablation	+4.05	6.43	−0.59
Flag-as-feature	2.87	360	72.0	Flag-as-feature	+4.01	6.09	+0.43
Plain	2.85	357	71.5	Zero-effect	+2.24	6.12	n/a

Table 1: Left: accuracy on 730 unseen days. Right: error on the *effect* of the appeal (percentage points of load) on ten held-out alert days; all four bootstrap intervals overlap, so no model separates from the zero-effect comparator. ρ_s is the Spearman correlation between implied and observed per-day effect, * $p = 0.011$.

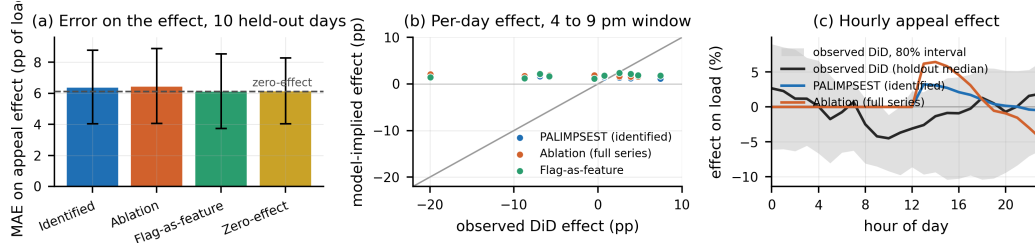


Figure 1: Ten held-out Flex Alert days, 2022-08-31 to 2022-09-09. (a) MAE on the appeal effect with bootstrap intervals; all four overlap and none separates from the zero-effect comparator (dashed). (b) Implied against observed per-day effect: observed spans +7.4 to −19.9 points while every model sits in a band one point wide. (c) Hourly appeal coefficients against the observed difference-in-differences profile with its 80 percent interval; the two identification strategies disagree twofold with each other and in sign with the holdout through the afternoon.

falls inside their envelope and what fraction of the query lies outside the observed support, which needs no ground truth and runs at serving time.

3 Results

Accuracy cannot tell the two identification strategies apart. In Table 1(A) the identified and ablation variants differ by 0.04 MAPE points, 5 units of pinball loss and 0.8 points of coverage, well inside run-to-run variation, so a practitioner choosing between a matched design and a naive regression sees no signal. Both lose to the plain model by roughly 0.9 MAPE points, and every model undercovers, with nominal 80 percent intervals containing 65 to 72 percent of hours. For scale, a 168-hour persistence baseline reaches 6.82 MAPE and a similar-day baseline 6.08. These six rows are equally compatible with the identified model being right about the causal structure, wrong about it, or carrying no causal information at all [Shmueli, 2010].

On the interventional holdout, no model separates from predicting nothing. Across the ten held-out days the observed appeal effect has mean −2.24 and standard deviation 8.19 percentage points. The identified model’s implied effect has standard deviation 0.17, about 2 percent of the observed dispersion, and never leaves [+1.16, +1.72]. Its mean absolute error on the effect is 6.36 with a bootstrap interval of [4.05, 8.78]; the ablation reaches 6.43, flag-as-feature 6.09, and predicting a zero effect every day reaches 6.12 (Table 1(B), Figure 1). Every interval overlaps every other, so the supported reading is that none of these models improves on the zero-effect comparator and both structural variants are nominally worse than it. On the quantity the behavioral layer exists to produce, the structure earns nothing.

Two details make this sharper than a wide error bar. The sign is wrong: the identified coefficients say appeals raise midday load by about 3 percent, an artifact of residual confounding with heat, while the observed effect is negative through most of the held-out afternoons. And the *ordering* across days is inverted ($\rho_s = -0.76$, $p = 0.011$), so a user asking which day the appeal worked best on would be steered toward the days it worked worst.

Two readings of the failure. Over the appeal window the 33 training alert days show a mean effect of +1.09 points (sd 4.15); the ten held-out days show -2.24 (sd 8.19), so the mean flips sign and the dispersion doubles. The 2022 heat wave carried an escalating alert sequence, demand response dispatch, and on 6 September the first statewide Wireless Emergency Alert for conservation [U.S. Energy Information Administration, 2022, California Public Advocates Office, 2022]. This admits two readings our design cannot separate: either the model fails to generalize an intervention effect out of sample, or it is *mis-asked*, because the binary appeal indicator encodes neither escalation, communication channel nor concurrent dispatch, so the 2022 treatment is not representable in its input. We think the second is at least partly right, and a richer treatment encoding is the obvious next experiment. That does not rescue the evaluation point, since under both readings the accuracy holdout certified a model whose interventional output was wrong, but it does move the remedy toward how the intervention is represented rather than how its coefficient is estimated.

Where structure does pay: the lockdown holdout. Holdout (C) gives the opposite verdict on structure and the same one on identification. Stripping the occupancy term out of the structural layer moves MAPE from 5.13 to 10.02 and turns a 0.8 percent bias into a 9.4 percent overforecast, because a model without it predicts a normal May. The term is worth 4.9 MAPE points, and the full hybrid at 2.55 (identified) and 2.36 (ablation) beats flag-as-feature (3.23) and plain (3.59), the one holdout where the structural model wins outright. Identification still buys nothing: the ablation is marginally better on both.

Replication and a serving-time diagnostic. Holdouts (D) and (E) are reported in full in Appendix B. Two headline points: under real randomization in Low Carbon London, two random halves of the same treated population show price responses of opposite sign through much of the day, so heterogeneity swamps the average effect the structural layer carries [Wolak, 2011]; and the analog envelope check disagrees with the model’s own counterfactual on 61 percent of hours even at $\Delta T = 0$, where no extrapolation is required, a warning computable without any held-out intervention.

4 What we think should be reported

Score the effect, and report a zero-effect comparator. Every counterfactual claim here survived the accuracy holdout and failed the interventional one. The accuracy table is compatible with the behavioral layer being right, wrong, or empty, and only the interventional holdout resolves it; a zero-effect comparator makes the null legible at the cost of one table row, and nothing beat it. Leakage-aware splitting, contamination audits and long-horizon consistency checks operate on the observation stream and cannot see a change in the treatment, so the unit of holdout should be the event and its institutional context.

Represent the intervention, and allow abstention. A stable administrative label can hide a changing intervention, and a binary indicator guarantees the model cannot tell the two apart; reporting what the treatment encoding contains, and holding out an event where it is known to have shifted, would have surfaced this before the coefficient was used. Structure and identification should likewise be ablated separately, since they carry opposite verdicts here. Finally, a model that can decline a rollout as outside what it supports is more useful in a decision loop than one whose answers are marginally better on average. We report the analog check as a diagnostic rather than a validated gate: it fires where holdout (B) later confirms a failure, and its sensitivity and false-alarm rate are not established.

Limitations. The interventional holdout is ten consecutive days from one heat wave, so they are not ten independent events and Table 1(B) treats days as exchangeable when temporal dependence makes that optimistic. That no model separates from zero is a claim about this event; that accuracy is uninformative about effect quality rests on Table 1 alone. The evaluation target is itself a matched difference-in-differences estimate inheriting parallel trends [Goodman-Bacon, 2021, de Chaisemartin and D’Haultfœuille, 2020] under confounded timing. With 43 alert days and one binary indicator we cannot separate interventional generalization failure from treatment misspecification. We study one balancing authority and one appeal program, and evaluate no pretrained foundation model.

References

- California Public Advocates Office. Preliminary analysis of california’s resiliency during the september 2022 heat wave. Technical report, California Public Utilities Commission, 2022.
- Brantly Callaway and Pedro H. C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021.
- Clément de Chaisemartin and Xavier D’Haultfœuille. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996, 2020.
- Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277, 2021.
- Tao Hong and Shu Fan. Probabilistic electric load forecasting: A tutorial review. *International Journal of Forecasting*, 32(3):914–938, 2016.
- Tao Hong, Pierre Pinson, Shu Fan, Hamidreza Zareipour, Alberto Troccoli, and Rob J. Hyndman. Probabilistic energy forecasting: Global energy forecasting competition 2014 and beyond. *International Journal of Forecasting*, 32(3):896–913, 2016.
- Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences*. Cambridge University Press, 2015.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, 2017.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.
- James Schofield, Richard Carmichael, Simon Tindemans, Mark Bilton, Matt Woolf, and Goran Strbac. Low Carbon London project: Data from the dynamic time-of-use electricity pricing trial, 2013. UK Data Service, SN 7857, 2015.
- Galit Shmueli. To explain or to predict? *Statistical Science*, 25(3):289–310, 2010.
- U.S. Energy Information Administration. California consumers respond to appeals for electricity conservation during heat wave. Today in Energy, 28 September 2022, 2022.
- Frank A. Wolak. Do residential customers respond to hourly prices? evidence from a dynamic pricing experiment. *American Economic Review: Papers and Proceedings*, 101(3):83–87, 2011.

A Model and data specification

This appendix gives the detail that Section 2 compresses and is written to be sufficient for reimplementation. All data sources are public: CAISO system load through OASIS, weather from ASOS station records, and the Low Carbon London trial from the UK Data Service, study 7857 [Schofield et al., 2015]. Code, the derived daily feature table, all result tables and the scripts that regenerate every figure are released under an open licence.

Task and data. The prediction target is hourly CAISO system load in MW, $y_{d,h}$ for day d and hour $h \in \{0, \dots, 23\}$, from 2016-01-01 to 2024-12-31 (3,287 days, 78,888 hours). Exogenous inputs are hourly dry-bulb temperature and dew point from public ASOS station records aggregated over the balancing authority footprint. Derived daily covariates are $T_{\max}$, $T_{\min}$, T_{mean} , a lagged afternoon temperature index $\tilde{T}_d$, day type (0 = weekday, 1 = Saturday, 2 = Sunday or federal holiday), day of year, and year. Two binary event indicators are attached per day: A_d for a CAISO Flex Alert (43 days, 2016 to 2022) and K_d for the pandemic occupancy period (89 days, 2020-03-19 to 2020-06-15).

Layer 1, calendar and thermal. The structural prediction is additive in MW:

$$s_{d,h} = B(\text{daytype}_d, \text{doy}_d, \text{year}_d, h) + \Theta(\tilde{T}_d),$$

where B is a smooth calendar baseline and the thermal response is piecewise in the smoothed index,

$$\Theta(\tilde{T}) = b_c (\tilde{T} - T_c)_+ + b_{c2} (\tilde{T} - T_c)_+^2 + b_h (T_h - \tilde{T})_+,$$

with $\tilde{T}_d$ an exponentially weighted afternoon temperature with retention α , so that $\tilde{T}_d = \alpha \tilde{T}_{d-1} + (1 - \alpha) T_d^{\text{pm}}$. The five thermal parameters and α are fitted by least squares on the training period. Fitted values under the two variants are given in Table 2; they agree to within 2 percent on every parameter, which is the basis for the claim that the two variants differ only in the behavioral layer.

Fit	α	T_c (°C)	T_h (°C)	b_c (MW)	b_{c2} (MW)	b_h (MW)
Identified	0.8472	17.654	13.235	1055.68	42.96	256.54
Ablation	0.8403	17.634	13.391	1073.97	39.61	257.63

Table 2: Fitted thermal parameters under the two identification strategies.

Layer 2, behavioral. Event effects enter as hour-of-day log-multiplicative coefficients applied to the structural prediction:

$$\hat{s}_{d,h} = s_{d,h} \cdot \exp(\beta_h^A A_d + \beta_{h, \text{daytype}_d}^K K_d).$$

The appeal coefficients β_h^A are constrained to zero outside hours 13 to 23, the window in which the Flex Alert appeal falls. The occupancy coefficients $\beta_{h,j}^K$ are estimated separately for each of the three day types, giving 72 coefficients. This is the *only* place the two variants differ.

Identification of Layer 2. For the *identified* variant, each event day d is matched to a pool of non-event control days drawn from a window of ± 45 days of year across all training years, selected on nearest $\tilde{T}$ and equal day type. The hourly estimate is the log-ratio contrast

$$\hat{\beta}_h = \frac{1}{|\mathcal{E}|} \sum_{d \in \mathcal{E}} \left[\log y_{d,h} - \log \bar{y}_{d,h}^{\text{ctrl}} \right] - \left[\log y_{d,h_0} - \log \bar{y}_{d,h_0}^{\text{ctrl}} \right],$$

differenced against a pre-window reference hour h_0 to remove day-level scale, then pooled over event days $\mathcal{E}$. This is the matched difference-in-differences design, and the identical procedure produces the “observed” effect on the held-out days in holdout (B). For the *ablation* variant, the same β are obtained by ordinary least squares of the log residual on the event indicator interacted with hour, using the full training series and no control-day matching.

Layer 3, residual. Gradient-boosted trees [Ke et al., 2017] are fit on the residual $y_{d,h} - \hat{s}_{d,h}$ with the pinball loss at $\tau \in \{0.1, 0.5, 0.9\}$, giving the quantile band. Features are: hour of day; day of week; month; day of year as sine and cosine; year; is-holiday; T and dew point at hour h and at lags 1, 2, 3 and 24 hours; daily $T_{\max}$, $T_{\min}$, T_{mean} and $\tilde{T}$; cooling and heating degree terms $(\tilde{T} - T_c)_+$ and $(T_h - \tilde{T})_+$; load at lags 24, 48 and 168 hours; and rolling 24-hour and 168-hour mean load. Event indicators are *excluded* from the residual model in both structural variants, so that all event information flows through Layer 2. Hyperparameters are 800 trees, learning rate 0.05, 31 leaves, feature fraction 0.9, with early stopping on a chronological validation tail of the training period.

Baselines. *Flag-as-feature* is the same gradient-boosted model fit directly on $y_{d,h}$ with A_d and K_d appended to the feature list and no structural layers; its counterfactual is obtained by setting $A_d = 0$ and re-predicting. *Plain* is identical but never receives A_d or K_d , so toggling the appeal changes nothing and its implied effect is exactly zero by construction. *Persistence* predicts $y_{d,h}$ by the value 168 hours earlier. *Similar-day* averages the $k = 12$ nearest training days in the standardized space of $(\tilde{T}, T_{\max}, \text{daytype}, \text{doy})$.

Counterfactual queries and the analog check. A query $\text{do}(\tilde{T} += \Delta, A = 1)$ recomputes Layers 1 and 2 with the modified inputs, holds the residual model’s non-thermal inputs fixed, and returns the change decomposed into thermal, behavioral and residual parts. The analog check retrieves the same $k = 12$ nearest days, forms the pointwise 10th to 90th percentile envelope of their normalized load shapes, and reports the fraction of counterfactual hours falling inside it, together with the fraction of query hours whose $\tilde{T}$ lies outside the observed training support.

Splits. The accuracy holdout is all of 2023 and 2024 (730 days). The lockdown holdout is 2020-05-01 to 2020-06-15 weekdays (46 days). The interventional holdout is 2022-08-31 to 2022-09-09 (10 days). These 786 days are excluded from every fitting stage, including the thermal least squares, the behavioral identification, and the residual model. The remaining 2,501 days are training.

B Holdouts (D) and (E) in full

(D) Low Carbon London. The trial placed 1,122 of 5,567 metered London households on a dynamic time-of-use tariff through 2013 [Schofield et al., 2015]. We split the treated households at random into halves A and B, identify the hourly price response on half A against the untreated control group, and apply it to predict half B. Carrying the identified response over gives 24.79 MAPE against 27.73 for a counterfactual with no behavioral term, a 2.9 point gain overall. During peak hours the behavioral term makes things *worse*, 26.15 against 23.99. Inspecting the halves directly, the response identified on A and the response observed on B differ in sign through much of the day despite random assignment. The practical reading is that between-household heterogeneity swamps the average treatment effect a single hour-of-day coefficient can carry, and that it does so in exactly the hours a system operator cares about. This analysis uses one random split; repeated splits with uncertainty estimates and an explicit heterogeneity decomposition would strengthen it and are the natural extension.

(E) Analog agreement. For a counterfactual query we retrieve the twelve nearest analog days and report three quantities: the fraction of query hours whose smoothed temperature lies outside the observed training support (*extrapolation*), the percentage of counterfactual hours falling inside the pointwise 10th to 90th percentile envelope of the analog days (*inside*), and the mean width of that envelope (*spread*).

ΔT	Extrapolation	Inside envelope (%)	Analog spread (%)	Mean analog distance
-3°C	0.40	44.5	4.54	1.32
0°C	0.00	39.1	4.45	0.98
$+3^\circ\text{C}$	0.48	37.4	4.46	1.36
$+6^\circ\text{C}$	0.90	38.0	4.88	2.37

Table 3: Analog agreement as the temperature intervention grows.

The extrapolation column behaves as expected: 0.00 at $\Delta T = 0$, rising to 0.90 at $+6^\circ\text{C}$, with mean distance to the nearest analog day rising from 0.98 to 2.37. That supports a serving-time refusal rule on large queries. The informative column is *inside*. Even at $\Delta T = 0$, where the counterfactual is the factual and no extrapolation is required, only 39.1 percent of hours fall inside the envelope against 4.5 percent mean spread, and the figure never exceeds 45 percent across the four query sizes. The model’s counterfactual disagrees with what similar historical days did on queries it should find easy. We emphasize that this is a diagnostic and not a validated gate: it fires in the one setting where holdout (B) later confirms a failure, and establishing that it is predictive of counterfactual error would require several intervention settings.

C Changes from the reviewed version

For transparency we list the substantive changes made after review.

1. **Corrected the zero-effect claim.** The reviewed version stated that the zero-effect baseline attained the lowest error on holdout (B). Flag-as-feature has MAE 6.09 against 6.12 for

the zero-effect baseline, so that ranking was wrong. The abstract, Table 1, Figure 1 and Section 3 now state that no model separates from the zero-effect comparator and that both structural variants are nominally worse than it, which is what the numbers support.

2. **Narrowed the framing.** The reviewed version was motivated by temporal foundation models and world models. We evaluate a task-specific structural model, so the abstract and a new Scope paragraph say this explicitly, and the limitations state that no pretrained temporal foundation model was evaluated.
3. **Added the treatment-misspecification reading.** A new paragraph in Section 3 sets out the two readings of the failure and states that our design cannot separate them.
4. **Clarified novelty.** A new paragraph in Section 2 distinguishes the established predictive-versus-causal-validity principle from the four specific contributions we claim.
5. **Regenerated Figure 1.** The reviewed version contained character-encoding errors in axis and panel labels. The figure has been rebuilt from the result tables with corrected labels.
6. **Added Appendices A and B.** The functional form, feature list, identification estimator, hyperparameters and splits are now specified, and holdouts (D) and (E) are reported in full with their numbers.
7. **Softened the trust-region claim.** Section 4 now reports the analog check as an unvalidated diagnostic and states what would be needed to establish it as a gate.